

An Automatic Virtual Makeup Scheme Based on Personal Color Analysis

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ABSTRACT

With people's timeless interest in beauty and the recent development of IT technologies, there has been a growing demand for a system that can suggest appropriate cosmetics or fashion styles to each individual. So far, professionals have made such suggestion subjectively or depending on personal experience, which produced inconsistent results. One possible method to do this objectively and systematically would be to utilize the personal color of an individual. Personal color, also known as skin tone color matching is the process of finding colors of clothing and makeup to match a person's skin complexion, eye color, and hair color. In this paper, we propose a scheme for automatically generating virtual makeup based on personal color. To do that, we first identify user's personal color from the image by analyzing the color of pupils, hair, and skin, and then select cosmetic method and colors. Lastly, by looking up the predefined makeup database for foundation, blush, lipstick, eyeliner, and eyeshadow, we produce a virtual makeup on the user's face image.

CCS CONCEPTS

• Information systems ~ Multimedia content creation

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KEYWORDS

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1 INTRODUCTION

It has been one of the common desires of all to make themselves look beautiful. Two most popular and long-lasting efforts to satisfy this desire is to find cosmetics and fashion styles that suit well. Recently, the concept of personal color has been developed in the cosmetics and fashion industry to determine the colors that best suit an individual's natural coloring. Personal color, also called skin tone color matching, color analysis, or seasonal color, is the process of finding colors of clothing and makeup to match a person's skin complexion, eye color, and hair color. Among a wide variety of approaches to analyze personal color, the most popular one is "seasonal" color analysis, which places individual coloring into four general categories: Winter, spring, summer and autumn. More recent systems subdivide the seasons into 12 or 16 categories. Personal color demonstrates how colors are capable of being flattering or, conversely, unflattering. For instance, colors that are unsuitable for the individual can make a person look pale or have flaws as wrinkles or uneven skin tone more noticeable. Nevertheless, they have been several problems in using personal

color widely. First, personal color has been costly and time-consuming since the user should visit professional diagnostic institutions and consult an expert such as colorist. Second, there is no standard training or degree required to market oneself as a color analyst. Color analysis is a marketing ploy that has been controversial since its beginnings. Third, personal color has been determined subjectively depending on personal experience. On the other hand, with the development of IT technologies, various smartphone applications for virtual makeup have been developed. Through these applications, users can make fun of their own face by putting on various make-ups virtually or figure out what type of makeup is right for them. However, one common problem with most of these applications is that the user should select many things such as color and style for makeup, which is very inconvenient to the unfamiliar user. Due to this, there has been a growing demand for an application that can recommend colors and makeup styles that suit the user and show the results of makeup virtually. Therefore, in this paper, we propose a method for recommending cosmetic methods based on personal color extracted from user selfie and making a virtual makeup. Section 2 introduces several related works including personal color analysis and virtual makeup for personal color analysis, and our proposed method is described in Section 3. In Section 4, we evaluate our proposed method through experiments and we conclude our paper in Section 5.

2 RELATED WORKS

So far, many works have been done to implement attractive makeup. Osborn et al. examined the beauty of human as biological structural trait [1]. They analyzed relationship between attractiveness and average face and found out that posture influence attractiveness judgments. Yael et al. proposed a technique for predicting facial attractiveness using machine learning technique for beauty ratings of facial images [2].

To generate virtual makeup that suits the selfie automatically, we should address two main problems. The first one is how to analyze personal color and the second one is how to put on virtual

makeup over the selfie. Before we describe our method in detail, we first introduce some related works that addressed personal color analysis and virtual makeup

2.1 Personal Color

Personal color began when Johannes Itten, a German professor of color science, confirmed that the color of a person's body was related to the color of the four seasons of nature [3]. He proposed a color analysis method that divides individual skin color into four seasons. American researcher Robert Doer studied coloring principles and published a color key system based on the evidence that skin tone has two color tones. Carole Jackson classified human images into four categories according to the season and suggested fashion and makeup through the color palette [4]. Ji-Ho et al. proposed a method for constructing a systematic color selection system using a personal color database and fuzzy logic [5]. They defined RGB color of skin color, hair color, eyes color and wrist color by classifying into four seasons and diagnosed personal color by using distance of user's body color and defined color in feature space.

2.2 Virtual Makeup

Oztel et al. proposed a lipstick coloring technique on users' lip [6]. It first extracts the lip region approximately by using face segmentation on users' image. Then, it converts color domain RGB to HSV and performs color segmentation for finding exact lip region. Finally, it applies various colors on lip. Guo et al. performed virtual makeup on input image based on example images, taken from a professional makeup book [7]. They used Active Shape Model (ASM) to extract facial landmark coordinates. Then, they performed matching face line between example image and user image. Next, they decomposed the images into three layers: face structure layer, skin detail layer and color layer. Then, they transferred the information of an example image to the user image and performed virtual makeup using the transferred information. Xu et al. improved a method proposed by Guo et al. [8]. In [7], when applying ASM algorithm, the position of each facial

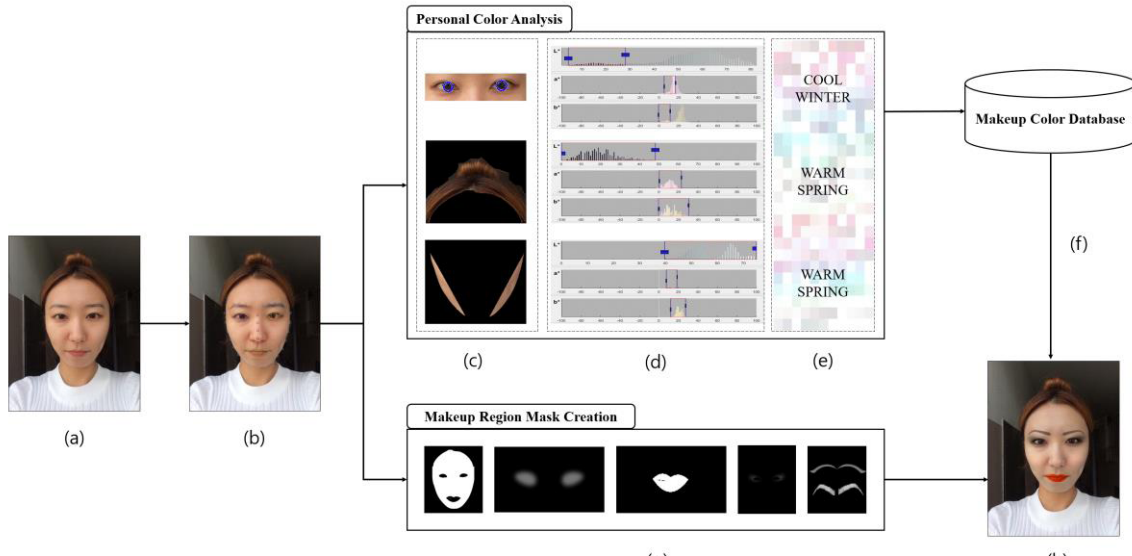


Figure 1: Overall steps of our proposed method.

landmark should be set manually. However, in this work, they used the Gaussian mixed models to automatically determine the skin region and extract the facial landmarks. Then, they performed virtual makeup after removing the elements such as wrinkles and scars by using Bilateral filtering.

3 PROPOSED METHOD

Our proposed method consists of two parts: Personal color analysis and virtual makeup generator. Fig. 1 shows the overall steps of our method. For an original input image in (a), we first extract the facial landmarks using Dlib [9]. Then, we calculate detailed coordinates of iris, hair, and skin from the landmarks, which are described in (c). After extracting detailed regions, we analyze the color of hair, skin and iris and based on this analysis, we determine its personal color. The landmarks are also used for establishing regions for makeup. Each region is 1-channel image that has same size as original image and describes specific area for makeup. We call it as 'makeup mask' and an example is shown in (g). After analyzing landmarks and creating masks, we perform virtual makeup. Colors for makeup are determined by taking into account the personal color. The relationship between personal color and its optimized color is predefined in the makeup database described in (f). A detailed description of each process is given in the next section.

3.1 Facial Landmark Extraction

To perform virtual makeup based on the personal color, we first extract the facial landmarks. To do that, we used the Dlib. Dlib is a machine learning algorithm implemented in C++. It is an open library and can be applied to various platforms. There is a pre-trained face marker in the Dlib, which allows you to extract 68 coordinates that map to the face structure. Fig. 2 shows an example of 68 landmark coordinates extracted by Dlib. The extracted face landmarks are used several times; we define it as *DFL* (Dlib Facial Landmark).

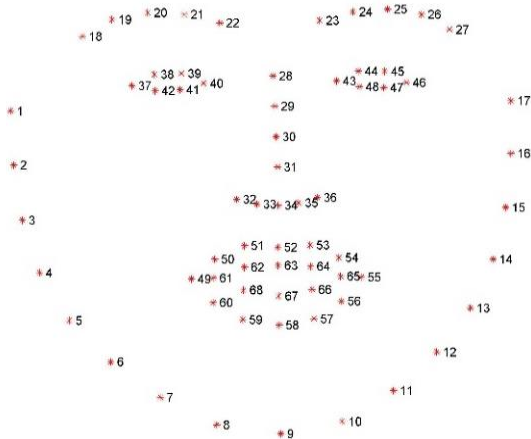
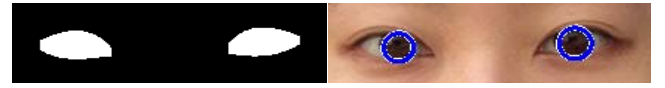


Figure 2: 68 points from dlib's facial landmark.

3.2 Personal Color Analysis

In order to analyze the personal color of an image, we use the iris, hair and skin color of the face. Thus, we first need to extract the detailed body regions needed for personal color analysis based on the *DFL*.

3.2.1 Iris color extraction. To detect iris, we first find candidates of eye regions in which the iris can exist. Each eye region extracted through the Dlib consists of a total of 12 coordinates. Generally, the human eye has an elliptical shape and hence we can define two different curves: the upper curve and the lower curve. To obtain two curves accurately, we applied 1d interpolation (1-dimension interpolation) to the upper curve and the lower curve [10]. Correct eye region can be extracted through these two curves.



(a) Eye mask (b) Pupil detection

Figure 3: An example of pupil detection.

Fig. 3(a) shows an example of extracted eye regions. Since the pupil in the eye has a circular shape, we can extract it using the Circle Hough Transform (CHT) that can find circles in the image [11]. The result of pupil detection by using Hough circle is described in Fig. 3(b). After detecting the pupil area, we extract the iris region for reliable personal color analysis. An iris is defined as the remaining region excluding the relatively dark region in the pupil. K-means clustering method was used to extract the color region having relatively high RGB values [12].

3.2.2 Hair color extraction. In general, the hair starts at the top of the head. We use this top region of the head to extract the original hair color. We first set the top of the head using the relative position of the landmark. Since the parietal region exists at a position higher than all positions of the facial landmarks, the region above the eyebrow which has the highest position among all landmarks, is defined as a candidate of hair region.

An example of such candidate region is shown in Fig. 4(a). In the case of hair color, various colors are possible due to many reasons such as race, artificial dyeing, etc. Since it is not easy to distinguish head and forehead with grayscale or binary images, we need to consider the overall color. We applied canny edge detection to the regions of R, G, and B, separately for extracting the hair region [13]. Fig. 4(b) shows the sum of the edges obtained from R, G, and B, and then, the morphological closing is applied to convert the set of lines into a binary image represented by the hair region and the rest region. The modified image called hair mask can be seen in Fig. 4(c). Since the white part in the figure represents the hair region, the color in this region is used for personal color analysis. The extracted hair region is shown in Fig. 4(d).

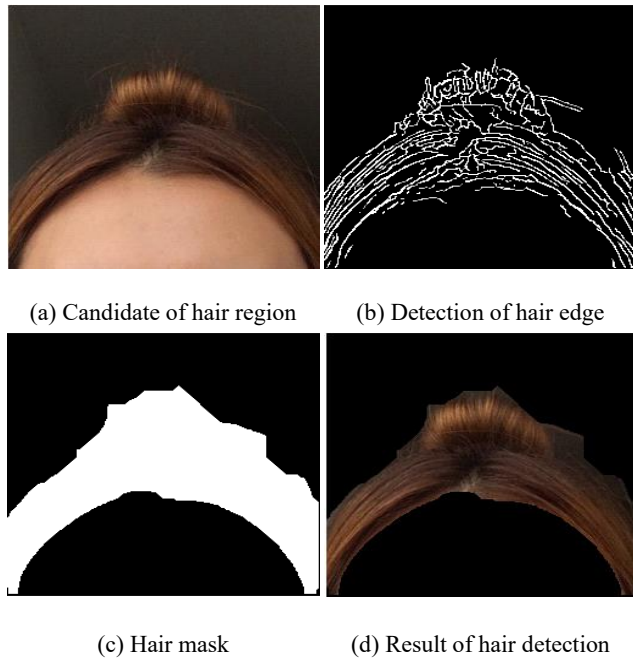


Figure 4: An example of hair region detection.

3.2.3 Skin color extraction. In order to analyze personal color, we use the region inside the cheek, near the jaw line. Since this part is relatively less influenced by sunlight, it preserves the original skin color well. Fig. 5 shows skin color regions used for personal color analysis. To extract the region, we apply 1d interpolation to jaw coordinates obtained from *DFL*. Then, we connect two points of two pairs as shown in Fig. 5.

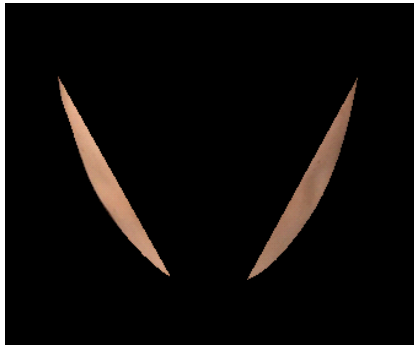


Figure 5: An example of skin region detection.

3.2.4 Personal color analysis. Based on the colors of pupils, hair, and skin, we can determine the personal color. Personal color is divided into two types; cool tone and warm tone. To decide the personal color, its RGB domain is converted into Lab domain. The tone of personal color is determined by 'a' and 'b' values of the skin region. The makeup color changes according to the skin color, thus it plays an important role in analyzing personal color. 'a' and 'b' are values that range from -128 to 127, and 'a' indicates the degree of red (+) and green (-), and 'b' indicates the degree of yellow (+) and blue (-) [14]. In general, if the value of 'a' is larger than the

value of 'b', it is classified as a cool tone and warm tone in the opposite case.

Cool tone and warm tone are again classified into four categories depending on the brightness of color: Spring, summer, autumn, and winter. In short, spring and autumn belong to warm tone and summer and winter belong to cool tone. In the same tone, spring / winter is brighter than autumn / summer. To distinguish them, we use the 'L' value in Lab domain, where 'L' has a value that range from 0 to 100. We determine seasonal type of personal color using the contrast of three bodily regions. Spring / winter and autumn / winter are separated through the experimentally obtained average contrast value '13'.

In order to perform a satisfactory virtual makeup, we also consider other information including user's age, gender, and purpose of makeup which can be obtained through the questionnaire. Finally, both colors that suit well for the user and makeup methods are retrieved from the predefined database based on the personal color and the questionnaire. The database contains diverse makeup methods defined by an expert such as colorist depending on colors and face landmarks.

3.3 Virtual Makeup

3.3.1 Foundation makeup. Foundation makeup is the first step in the ordinary cosmetic procedure and corresponds to the makeup base that makes skin tones look different overall. To apply the makeup base, we need to know the skin region from the whole face. Skin region can be extracted very quickly by using skin region explicitly defined by Equation (1) [15].

$$\begin{aligned} R > 95 \text{ and } G > 20 \text{ and } B > 20 \text{ and} \\ \max(R, G, B) - \min(R, G, B) > 15 \text{ and} \\ |R - G| > 15 \text{ and } R > G \text{ and } R > B \end{aligned} \quad (1)$$

However, even if this condition is satisfied, the hair color, background, clothes and the like could be similar to the color of the skin, and they may be recognized as a skin region in some cases. To solve this problem, we limit the facial skin region as follows. We crop the outline according to the facial shape and remove the landmarks inside the face to extract the correct face region. The upper limit part of the face is the boundary between forehead and hair and the lower limit is the jaw line obtained by *DFL*. After removing outer parts of the face, 1d interpolation is applied to eyes and lips coordinates inside the face to find the whole coordinates of those regions, and the regions are removed. Fig. 6 shows an example of the extracted face region. The white area is the region that makeup foundation should be made.



Figure 6: Mask of skin.

After accurately detecting facial skin region, its color is changed into foundation color. It is important to maintain the naturalness of the skin while changing the tone of the skin so that we convert color domain from RGB to Lab. After conversion, the changed skin color can be obtained using Equation (2).

$$P_{out}(i, j) = \alpha \times \left\{ P_{in}(i, j) + Cosmetic_{lab} - \frac{1}{n} \sum_x \sum_y P_{in}(x, y) \right\} \quad (2)$$

Here, P represents Pixel and $0 < \alpha < 1$

Equation (2) is applied equally to each 'L', 'a', and 'b' channel, and P represents 'L', 'a', and 'b' values. x and y are the coordinates of a point in the face region. n is the total number of pixels in the face region. i and j represent the skin coordinates of the face region and α is a constant that indicates the strength of makeup. Since 'L', 'a' and 'b' values can exceed the maximum or minimum value, it is necessary to adjust these values. The final value of the face region whose range has been adjusted is shown in Equation (3).

$$P_{final}(i, j) = \begin{cases} limit_{min}, & P_{out}(i, j) < limit_{min} \\ limit_{max}, & P_{out}(i, j) > limit_{max} \\ P_{out}(i, j), & Others \end{cases} \quad (3)$$

Not only makeup foundation but also other color makeup methods use Equations (2) and (3) when applying cosmetic color on facial region.

3.3.2 Blush. Blush is a cheek makeup to give various effects, such as making the skin look vivid or complementing the cheekbones. To perform the blush makeup, it is necessary to extract the cheek region. Four major coordinates based on the *DFL* are selected for finding cheek region. After setting four points, we generate closed curves through applying 1d interpolation to these four points. Then, we extract all the coordinates inside the closed curves. In the case of blush, since it is expressed in both cheeks, the coordinates of the final cheek region are extracted by symmetrically aligning the cheek region coordinates extracted from the right cheek with reference to the center of the face. Next, we apply Gaussian blurring on cheek region to put gradation effect that gives blush a natural effect as shown in Fig. 7.

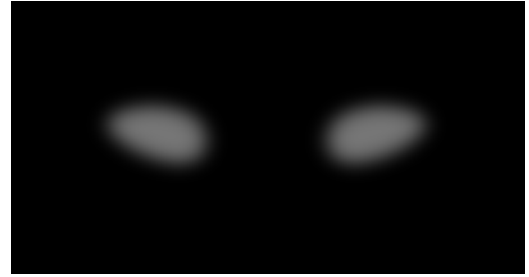
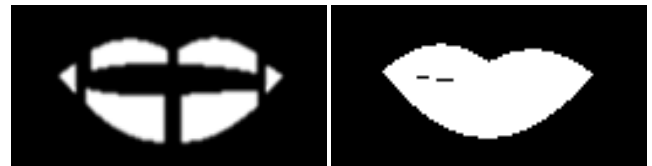


Figure 7: Mask of blush.

3.3.3 Lip. Lip makeup changes color and texture of lips. The lips region, which can be obtained by *DFL*, is divided into oral angle, upper lip, and lower lip as described in Fig. 8(a).

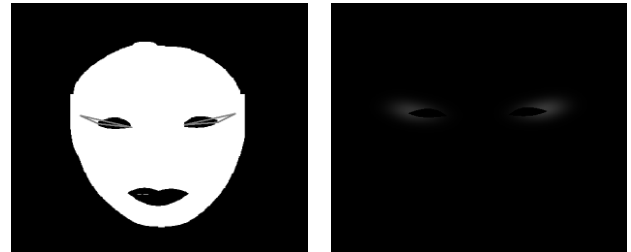


(a) Distributed lips region (b) Mask of lips

Figure 8: An example of lips region detection.

To extract exact lips line, we apply 1d-interpolation to each segmented region. Connecting all points forms lips shape which is shown in Fig. 8(b).

3.3.4 Eyeshadow. Eyeshadow puts color onto the eyelids. It is important to set exactly where to put on makeup. To generate a mask that represents the makeup region, we use a relative position between eye and eyebrows. We define 3~4 major points which are calculated by the relative position. For example, we set 3 major points as shown in Fig. 9(a). The first and last points are both end points of the upper line of eye and the second point is defined as 1/3 point between the end of the eyebrow and the end of the eye. Next, the created points are connected to generate a shadow region mask. The shape of the shadow depends on the makeup situation, so we predefined the shape according to the situation in the database. Fig. 9(b) is the result of applying Gauss blur in order to bestow similar effect to that applied on blush.



(a) Connected 3 major points (b) Mask of eye shadow

Figure 9: Example of eye shadow region creation.

3.3.5 Eyeline and Eyebrow. Eyeline and eyebrow are various shapes according to makeup style. Therefore, we performed virtual makeup by mapping predefined templates to the eye and eyebrow region. Since the sizes and positions of eye and eyebrows are

different according to the image, we performed preprocessing for accurate template mapping. In the case of eye lines, we measure the length of the eyes using the start and end coordinates of the upper line of the eye region obtained from *DFL*. Then, resized the template to fit the length and matched the coordinates of the template and top edge of the eye. In the case of eyebrows, the region of existing eyebrows is filled with surrounding skin color for a natural makeup. After that, template mapping is performed as same as method for eyeline. Fig. 10(a) shows the eyeline makeup result and Fig. 10(b) shows the eyebrow makeup result.



(a) Example of eyeline (b) Example of eyebrow

Figure 10: Example of template matching.

4 EXPERIMENTAL RESULTS

In order to evaluate the effectiveness of our proposed method, we perform two experiments. In the first experiment, we test how accurately facial landmarks such as iris, hair and jaw lines are detected for personal color analysis. In the second experiment, we measure how satisfied the user is with the results of the virtual makeup. To do this, we construct a data set. We took 5 face photos from each 20 persons at natural light environment thus, total 100 pictures are used for analyzing personal color and evaluate the satisfaction according to the result of virtual makeup. The photographs used in the experiment were taken using an iPhone7, and the experiments were performed on Matlab 2017a, Python 3.5, and OpenCV 2.4 environments.

4.1 Personal Color

The iris, hair, and skin region were extracted from the 100 photographs, and the extraction accuracy of each was as shown in Fig. 11. Overall accuracy is high because the basic facial landmarks are extracted using Dlib. Thus, if the accuracy of the library becomes higher, the feature extraction accuracy becomes higher. In comparison with the eye and jaw line, the detection accuracy of hair is relatively low. This is because the hair style and color are different compared to other parts so that the accuracy is low in case of unusual hair style or hair color similar to skin.

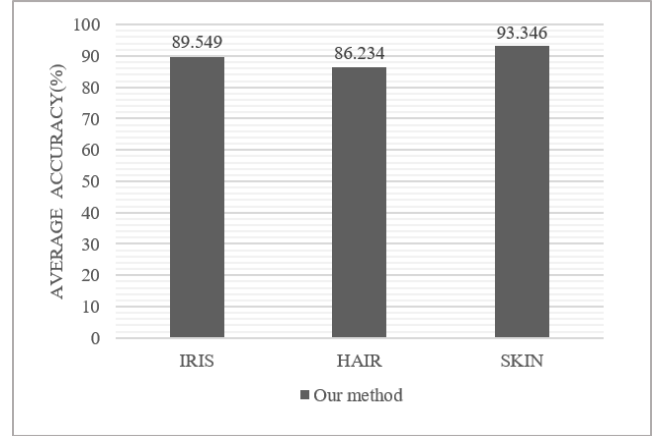
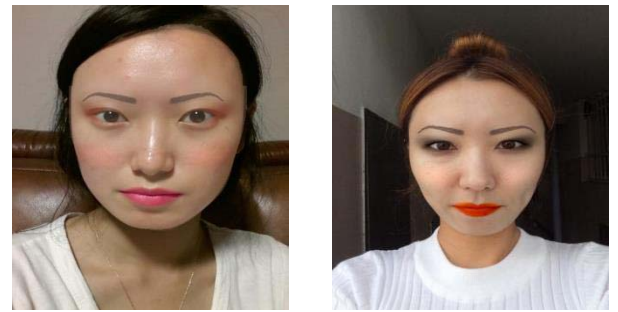


Figure 11: Accuracy of personal color feature extraction.

4.2 Virtual Makeup

To evaluate user satisfaction for virtual makeup, we investigated satisfaction of 20 subjects for their virtual makeup photograph. Fig. 12 shows an example of virtual makeup images used in the evaluation. Fig. 12 is virtual makeup result for two warm tone persons. Fig 12(a) shows warm tone-autumn type and is performed daily makeup. Fig 12(b) is warm tone-spring type, and is applied chic makeup. We could put virtual makeup on makeup foundation, lip, blush, eyeline, eyeshadow and eyebrow. To evaluate the user satisfaction for virtual makeup, we conducted a survey on five makeup items. The question items are as follows: 1) Selected makeup color, 2) Size of makeup region, 3) Location of makeup, 4) Intensity of makeup.



(a) Warm tone / Autumn (b) Warm tone / Spring

Figure 12: An example of virtual makeup.

Each item is rated 1~5, from very poor to very good. The result of satisfaction evaluation is shown in Fig. 13. Foundation, blush, and shadow show average satisfaction score of 4 or higher among five makeup method. However, lips, eyeline & eyebrow shows relatively low score. In the case of lips, it draws fixed shape based on extracted coordinates so that it has difference with real lips line in some case. For this case, it got low score in the 'size of makeup region'. Since we do not consider the curvature difference between template and eyeline/eyebrow, it shows low matching accuracy. When the curvature of the eyeline or eyebrow is significantly different from the curvature of the template.

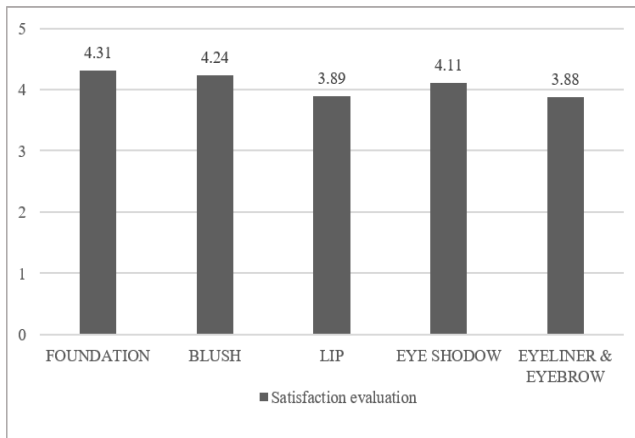


Figure 13: Average score for satisfaction evaluation.

5 CONCLUSIONS

In this paper, we proposed a scheme for virtual makeup based on personal color analysis. To do that, we first extracted the facial landmarks such as iris, hair and skin region using Dlib's pre-trained face detector. Then, we analyzed the color of the extracted regions and decided the personal color by considering landmark features and user information. Lastly, we performed virtual makeup by looking up the predefined makeup database that contains cosmetic methods and colors depending on the personal color and user situation. We evaluated our scheme through two experiments. The experimental results show that the accuracy of hair, skin and iris region extraction is quite good and user satisfaction for virtual makeup is satisfactory overall.

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